

Electrical Machines

BRANCH: ELECTRICAL (EE)

COURSE CODE: EE202

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Electrical Machines (EE202) under the Electrical (EE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Transformers

Single-phase transformers: working principle, EMF equation, equivalent circuit, phasor diagrams, losses, efficiency, voltage regulation. Three-phase transformers connections.

Unit 2: DC Machines

DC Generator: construction, armature winding, EMF equation, commutation, characteristics. DC Motor: Back EMF, speed control methods, starting, losses, efficiency.

Unit 3: Three-Phase Induction Motors

Production of rotating magnetic field, slip, equivalent circuit, torque-slip characteristics, starting methods, speed control.

Unit 4: Synchronous Machines

Synchronous Generator (Alternator): EMF equation, armature reaction, voltage regulation (EMF, MMF methods). Synchronous Motor: principle of operation, V-curves.

*EMF eq of Transformer: $E = 4.44 * f * N * \Phi$. Slip of Induction Motor: $s = (N_s - N_r) / N_s$.*

CRITICAL FORMULA / PRINCIPLE: Key Formulas & Exam Pointers

